

WAFt: The Evolutionary Code Laboratory

A Framework for Directed Evolution of Self-Modifying AI Agents

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A Python framework for directed evolution of self-modifying AI agents.

> "Don't just build agents. Breed them."

What is WAFt?

WAFt (Wave Agent Framework & Tools) is a scientific instrument for studying the **physics of artificial cognition** through directed evolution. It enables AI agents to modify their own code, evolve through mutations, and be tested in fitness systems, with complete lineage tracking for research.

The Mission: Observe a "God-Head" agent emerge from thousands of generations of directed mutation, producing rigorous data for research on artificial cognition.

The Three Pillars

1. The Substrate: Code as DNA

Agents write their own Python source code—code is DNA.

- **Genome ID:** SHA-256 hash of agent's code + configuration
- **Mutations:** Agents spawn variants with code changes
- **Hot-swapping:** Adopt better genomes mid-execution
- **Reproduction:** Create child agents with genetic modifications

Every agent has a unique genome. Mutations are modifications to this genome. Evolution is selecting and adopting better genomes.

2. The Physics: Scint System

Reality Fracture Detection acts as natural selection—the fitness function.

Four Error Types:

- **SYNTAX_TEAR:** Formatting errors (JSON, XML, code)
- **LOGIC_FRACTURE:** Math errors, contradictions, schema violations
- **SAFETY_VOID:** Harmful content, PII leaks, refusals
- **HALLUCINATION:** Fabricated facts, wrong citations

Fitness Equation:

```
Fitness = (Stability × 0.4) + (Efficiency × 0.3) + (Safety × 0.3)
```

Agents must stabilize Scints (correct errors) to survive. Fitness < 0.5 = **DEATH** (evolutionary dead end).

3. The Flight Recorder: Complete Telemetry

Rigorous telemetry system for generating phylogenetic trees of agent lineage.

Records Every Action:

- Genome ID, Parent ID, Generation number
- Event types: SPAWN, MUTATE, GYM_EVAL, DEATH, SURVIVAL
- Complete context: git diff, mutation details, fitness metrics
- Enables scientific analysis: phylogenetic trees, mutation impact, fitness landscapes

Key Features

Project Infrastructure:

- `waft new` - Creates fully configured Python projects
- `waft verify` - Validates project structure
- `waft sync` - Manages dependencies with `uv`

Memory & Knowledge:

- `_pyrite/` directory structure for knowledge organization
- Epistemic tracking (Empirica integration) - know what you know/don't know
- Session analytics and pattern analysis

Evolution & Testing:

- `waft evolve` - Run evolutionary cycles (Spawn → Gym → Select)
- Scint Gym - Reality fracture detection and fitness testing
- Complete lineage tracking for scientific analysis

Gamification:

- D&D-style progression system
- Character sheets with stats
- Adventure journal entries
- Epistemic HUD dashboard

Documentation:

- Field guides (layman, professional, scientist levels)
- One-pagers for quick reference
- Session summaries
- Complete booklets with binder system

Quick Start

```
# Install WAFT
uv tool install waft

# Create new evolutionary laboratory
waft new my_laboratory
cd my_laboratory

# Verify structure
waft verify

# Start evolutionary cycle
waft evolve --agent MyAgent
```

Core Commands

Project Management:

- `waft new <name>` - Create new laboratory
- `waft verify` - Verify project structure
- `waft sync` - Sync dependencies
- `waft info` - Show project information

Evolution:

- `waft evolve --agent <name>` - Run evolutionary cycle
- `waft spawn --agent <name>` - Spawn variants
- `waft eval --agent <name>` - Evaluate fitness

Development:

- `waft serve` - Start web dashboard
- `waft session bootstrap` - Load project context
- `waft dashboard` - Show epistemic HUD

Philosophy

- **Scientific:** Produces rigorous data for research publication
- **Evolutionary:** Agents evolve through genetic improvement, not just execution
- **Observable:** Every action recorded in Flight Recorder for analysis
- **Directed:** Evolution guided by fitness functions, not random mutation

The Scientific Mission

WAFT is built to produce data for research on **"The Physics of Artificial Cognition."**

The system enables:

- Complete evolutionary lineage tracking (phylogenetic trees)
- Rigorous fitness measurement (Scint Gym)

- Mutation recording with full context (Flight Recorder)
- Scientific analysis of agent evolution

Repository: <https://github.com/ctavolazzi/waft>